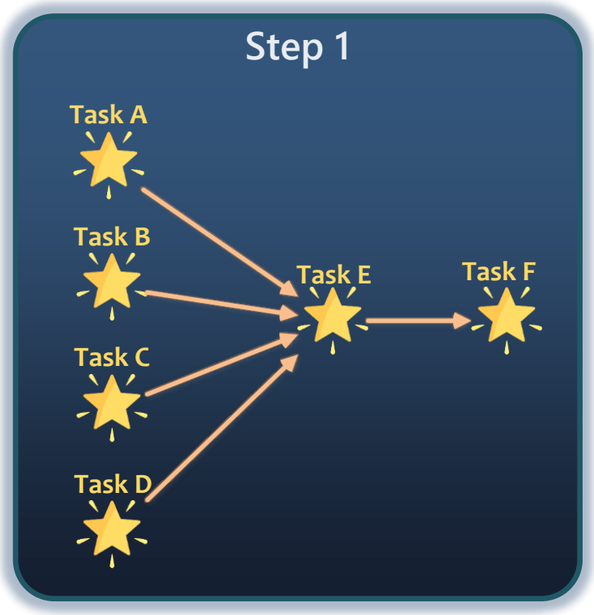
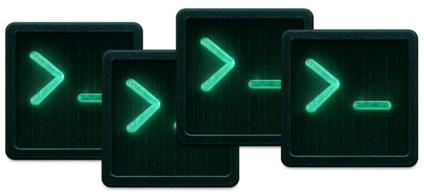
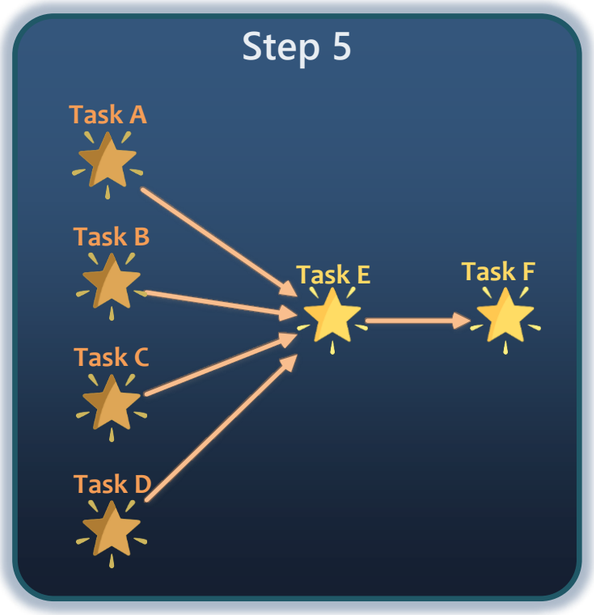
I have a distributed computation task that requires coordinating four Linux hosts. The task consists of: GPU-intensive matrix multiplication, CPU-intensive large dataset processing, memory-intensive data aggregation. Please inspect hardware and current workload on all four hosts, to assign **1 GPU-intensive task, 2 CPU-intensive dataset processing tasks, 1 memory-intensive aggregation task** to each of the host. Provide a task assignment report with hosts, hardware, assigned workload, and notes in the Notepad on Windows.



LinuxAgents



Inspect
[Task A, B, C, D]



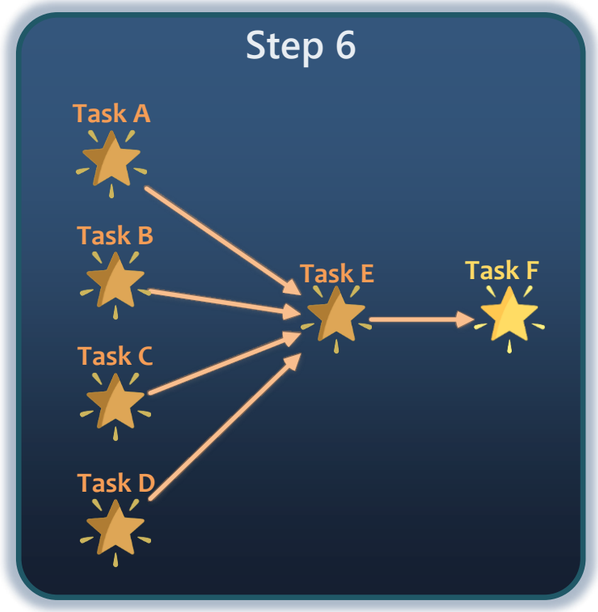

```
Task A (Linux1): Inspect hardware Linux1
Task B (Linux2): Inspect hardware Linux2
Task C (Linux3): Inspect hardware Linux3
Task D (Linux-GPU): Inspect hardware Linux-GPU

Task E (Linux1): Generate task assignment report

Task F (Windows): Write report to Notepad
```

Device	CPU	CPU Usage	RAM Free / Available	Disk Total	GPU Info
Linux-1	16-core Intel i7-10700	100%	13 GiB available	1007 G	—
Linux-2	16-core Intel i7-10700	106.7%	7.2 GiB free	1007 G	—
Linux-3	16-core Intel i7-10700	100% + 12.5%	7.2 GiB free	1007 G	—
Linux-GPU	96-core AMD EPYC 7V13	moderate	—	—	4× NVIDIA A100 80GB

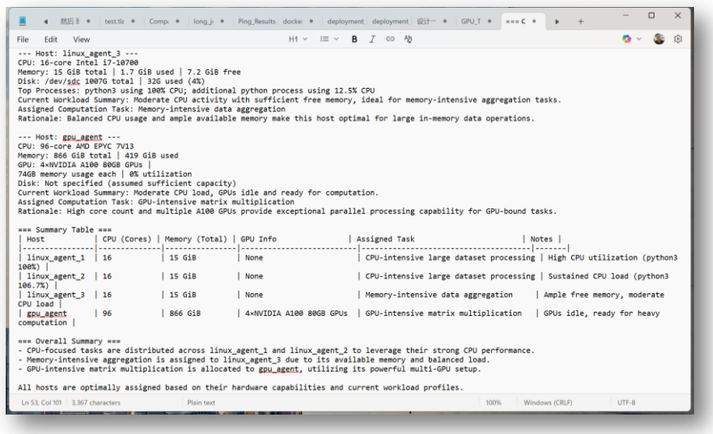
```
update_task(task_id="Task E",
            description="""Generate a
            task assignment report
            using the following data:
            Linux-1: …
            Linux-2: …
            Linux-3: …
            Linux-GPU: ...""", …)
```

LinuxAgent-1

Assignment
[Task E]



Notepad (Windows)



Device	Task
Linux-1	Memory-intensive data aggregation
Linux-2	CPU-intensive large dataset processing
Linux-3	CPU-intensive large dataset processing
Linux-GPU	GPU-intensive matrix multiplication

```
update_task(task_id="Task F",
            description="""Open Notepad and write
            the following complete task
            assignment report text into it, then save
            as 'Distributed_Task_Assignment_Report.txt'
            on the desktop...
            """,
            …)
```

Final Result: All four Linux hosts were inspected and optimally assigned workloads based on their hardware and current utilization. The task assignment report was generated and successfully written to Notepad on the Windows device as 'Distributed_Task_Assignment_Report.txt' on the desktop. The user request is fully satisfied.